

Forensic Analysis of Private Mode Browsing Artifacts in Portable Web Browsers Using Memory Forensics

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Abstract— The popularity of portable web browsers is increasing due to its convenient and compact nature along with the benefit of the data being stored and transferred easily using a USB drive. As technology gets updated frequently, developers are working on web browsers that can be portable in nature with additional security features like private mode browsing, built in ad blockers etc. The increased probability of using portable web browsers for carrying out nefarious activities is a result of cybercriminals with the thought that if they use portable web browsers in private mode it won't leave a digital footprint. Hence, the research paper aims at performing a comparative study of four portable web browsers namely Brave, TOR, Vivaldi, and Maxthon along with various memory acquisition tools to understand the quantity and quality of the data that can be recovered from the memory dump in two different conditions that is when the browser tabs were open and when the browser tabs were closed in a system to aid the forensic investigators.

Keywords— *Forensics, Memory, Portable, Private Mode, Web Browser*

I. INTRODUCTION

We are living in the era of technology and people need access to the internet for various purposes like online shopping, online payments, e-learning and many more, hence, web browsers are being used. Browsers are developed in a manner to store some of the user's activities for better assistance like caching files and images, web history, search history, cookies, user ID and password (If allowed) and others. Many individuals are becoming more aware of their digital footprints and demands for privacy as technology gets more incorporated into our daily lives. The purpose of privacy is to ensure that no organization or institution obtains or keeps any potentially sensitive information about users without their permission, such as web history, geolocation, account data, and so on. To maintain the privacy of the users, most of the browser companies have developed a feature called private browsing mode. It is stated that the private browsing window does not save any history, cookies, or files in its cache hence, user can surf the internet without leaving many traces on their local PC, such as sites visited, account details, download history, and so on. A portable browser is a browser that has been modified to use developer packages and can be run straight from a portable storage device like a USB flash drive. The key advantage of a portable web browser is that you don't have to install it on every device and that you can use all of your bookmarks, history, private mode, and add-ons without leaving any traces on the machine. All of the residual artifacts are saved on the portable device.

The use of such a feature can be beneficial to consumers in a variety of ways, but forensic investigators may find it difficult to acquire information and recreate events. The investigator may not always have access to the portable USB drive from which the portable browser was run, and the portable browser does not save much information on other storage devices. A memory dump contains valuable forensic data regarding the state of the system prior to an occurrence such as a crash or a security breach. Thus, memory analysis could be useful for examining the user activities with portable browser. Tools used for the memory acquisition have a significant impact on the investigation of such crucial cases. In such a circumstance, it is critical to identify the optimum memory acquisition tool. The goal of this study is to determine which artefacts were left behind after specific actions were performed in portable web browsers' private mode. Furthermore, to determine which memory acquisition tool produces the best results in the same situation.

II. MAJOR PORTABLE WEB BROWSERS WITH PRIVATE MODE FEATURE

The major portable browsers that have a private mode function are described in this section of the article. Perpetrators frequently employ portable browsers to secure their operations on a local system. All of the traces of portable web browsers are saved on USB drives. Private mode surfing is a browser feature that allows users to retain their privacy by deleting all traces of their browsing activities from the local system. The investigation was conducted on four popular browsers: Brave, TOR, Vivaldi, and Maxthon.

A. Brave

Brendan Eich created Brave, an open source and publicly downloadable browser, in 2015 [1]. Brave browser comes with an ad blocker by default, and it also blocks site tracker cookies, giving you more privacy when surfing the web. Because Brave does not have a portable version, it has to be downloaded from a different source [2].

B. TOR

TOR (The Onion Routing) is a free and open source browser for anonymous internet access. It can be used as a portable browser by extracting it directly from its archive onto a USB stick. When browsing the dark web, it leverages the Tor network to safeguard the user's privacy and anonymity. Furthermore, when used in portable mode, it does not save any traces of the user's activity on the local computer.

C. Vivaldi

Vivaldi browser was first released in 2016 based on chromium base code [3]. It is regarded as the most customised web browser, and its portable feature ensures more privacy. Other browser extensions, such as Google Chrome and Firefox, can be used with Vivaldi.

D. Maxthon

Maxthon is a freeware Chinese browser that is based on the chromium source code [4]. Maxthon's most recent version is well-known for its technical support for blockchain-based websites. It now includes new features such as UUMail, which filters out fraudulent emails, Maxnote, which keeps notes, and Passkeeper, which securely remembers passwords.

III. LITERATURE REVIEW

In the research article, Mahendrakar et al. [5] chose four browsers (Mozilla Firefox, Internet Explorer, Google Chrome, and Safari) and examined the artifacts left behind after utilizing private mode in each of them. For a controlled setting, they established their own website and interacted with it. They took a snapshot when the private browsing mode window was open and another immediately after it was closed. They discovered that until they are overwritten, entries in Page Descriptor Table Entries (PDEs) and Page Table Entries (PTEs) remain intact. All entries are zeroed when the private browsing window is closed. However, it can be collected using forensic tools by exploiting baseline memory space that was previously taken when the browsing window was open. They've also created a tool that wipes the memory region as soon as the operation is over, so it can't be restored.

Said et al. [6] have discovered traces left by three commonly used browsers' private mode (Firefox, Google Chrome and Internet Explorer). They looked for evidence in common files and folders where each browser's history and cache files are saved, but none were located. Then they looked at physical memory, and they were able to retrieve some of the data, such as typed URLs and keywords. Finally, they imaged the hard disk and examined pagefile.sys, a file that stores a copy of physical memory, where they discovered some private browser history trails. They come to the conclusion that most browsers' private sessions may be retrieved, however Internet Explorer saves browsing history and cache to the hard disk and then deletes them when the browser is closed, hence, that can be recovered and also opine that Firefox and Google Chrome are the best in terms of privacy.

Satvat et al. [7] have revealed flaws in popular browsers' private browsing mode. Local attacker and Remote attacker were the two threat models they chose. They developed a browser extension to explore the influence of extensions on private windows, and they advised that extensions be deactivated in private mode since they can access private browsing data.

Choi et al. [8] stated that the portable web browser activities can be identified by investigating certain windows features like prefetch which shows program execution even portable program and UserAssist key of windows registry.

Nelson et al. [9] used popular web browsers such as TOR, Google Chrome, and Mozilla Firefox, as well as their respective private modes, to extract and compare which forensic artifacts could be retrieved. They used FTK Toolkit,

FTK Registry viewer, WinHex, FTK imager and SQLite manager Firefox add-on. The end result indicates that the majority of the data was retrieved from standard browsing modes, with very little data recovered from private browsing and nearly no artifacts recovered from the TOR Browser.

Jadoon et al. [10] studied about TOR Browser to check its anonymity and privacy by performing forensic analysis of Tor on Windows 8.1. They observed windows registry, hard disk and physical memory for all the leftover artifacts that Tor browser transfers to the user's system when it's open and when it's closed. They searched for the artifacts about Tor installation, usage and browsing activities. Eventually results show that the Tor browser transfers many artifacts on user system especially in device's physical memory.

Muir et al. [11] evaluated the effectiveness of Tor Browser in protecting the user against such an adversary by performing a forensic examination of the software and its communication with the host operating system. It was carried out using virtualization and a pre-determined browsing protocol that allows artifact recovery with static and live forensic techniques, such as monitoring processes, searching for keywords and file carving, with the help of open-source digital forensic tools like Autopsy and the Volatility Framework. According to the findings, the TOR browser does not adequately protect its users.

Horsman et al. [12] analyzed thirty web browsers to identify the presence of a private mode artifacts. For the objective of open, transparent scrutiny and review from individuals working in this field, the approach was recorded, and the outcomes and limitations were discussed.

Mahlous et al. [13] comprehensively studied brave's private browsing mode for its privacy measures through forensic investigation. Authors have identified the location and type of evidence available through live and post-mortem state analysis. The research comprises of a series of experiments that disclose browser functionality and illustrate the usage of appropriate tools for residual artifact retrieval. The results revealed that, despite leaving no traces of browsing activity on the hard disc, visited URLs, photos, keyword searches, and even cached movies were recovered from the physical memory, indicating that the brave browser lacks the capacity to offer users with adequate privacy.

Gabet et al. [14] investigated three privacy-enhancing browsers, Dooble, Comodo Dragon, and Epic, as well as three widely used browsers, Chrome, Edge, and Firefox, in their private mode browsing window. Furthermore, they examined the results of two extensively utilized forensic tools, FTK and Autopsy. Finally, they came to the conclusion that browsers' claims of greater privacy are ineffective in terms of providing privacy to users. Firefox and Doodle, out of the six browsers tested, delivered the satisfactory performance in terms of privacy. They found that FTK can parse more results than Autopsy when it comes to forensic investigation tools.

The use of the Tor Browser, which is known for maintaining anonymity on the dark web, has been identified as a key problem in digital forensic analysis, so Sajan et al. [15] have performed an in-depth analysis of Tor browser to see what all information/artifacts can be retrieved from the system files and RAM dump. HxD, Volatility, DumpIT, and Win-LiFT were utilised to conduct the experiment. Finally, the study indicated that Tor browser features default encryption and other security policies that may be discovered through

memory analysis, offering significant information to the researchers.

Hughes et al. [16] investigate the use of private mode in the selected web browsers, namely Google Chrome, Microsoft Edge, Brave, and Mozilla Firefox, and compare the results to those of normal web browsing mode to determine which web browser offers the most privacy and whether the results correspond to what is supported by the web browser vendor. The tools used to conduct the experiment are FTK Imager, Bulk Extractor, Autopsy, and BrowsingHistoryView. According to the findings, Google Chrome has a superior incognito option than the other browsers, with Brave being the least private.

Hasan et al. [17] studied the privacy that web browsers provide in private mode in order to show a clear path to follow if a forensic analyst wanted to retrieve crime-related data from a machine that was suspected of being surfed on in that mode. Google Chrome, Mozilla Firefox, and Microsoft Edge were picked as the most popular Windows compatible browsers. Minitool Power Data Recovery tool, Process Monitor, and Browser history Examiner were employed in the research. The paper concludes by demonstrating that the privacy guaranteed in private mode varies by browser. In comparison to the other two browsers, Google Chrome performed better.

IV. EXPERIMENTAL SETUP

Aim of this research is to analyze the potential artifacts left over by the popular portable web browsers such as Brave, Vivaldi, Maxthon, and Tor in their respective private mode. Besides that, to check whether the web browser can actually maintain the privacy of the user using private mode of portable web browsers, and to compare the effectiveness of different Memory Forensic Tools. In order to get the realistic data for the experiments, it was carried out in a simulated environment that was created using virtualization technique. The virtual computers were just utilized for browsing exploration and were not used for anything else to avoid the interventions in the results. Data was gathered on known keyword and searching basis over the course of a long duration. The browser was closed, the virtual computer was shut down, and a snapshot image of the virtual machine was saved after a browsing session concluded. The surfing and data storage steps were equivalent among browsers and virtual computers. Table I shows the tools and versions used for the experiment.

TABLE I. NAME AND VERSIONS OF THE TOOL USED FOR EXPERIMENT

Name of the tool	Version
VMWare	16.2.1
Brave	1.20.110-71
Tor	10.5.2
Maxthon	6.1.2000
Vivaldi	3.7.2218.55
Dumplt	1.3.2.20110401
Magnet RAM Capture	1.20
Belkasoft Live Ram Capturer	1.0
Volatility Framework	2.6.1

Begin the online experience by connecting the portable web browser to the VM and running it in private mode, then looking for Facebook and Gmail, signing in to the following sites, and sending an email with Gmail. Then go to YouTube and look for a video on how to crack a password. Go to amazon.com and search for Samsung Galaxy Note 10 Lite in the newest mobile phones section. Download an image of a

cybercrime in jpg format. After completing all of these tasks on the browsers, memory was acquired in two scenarios: when the tabs were open and while they were closed, after the browsing session had ended. To reduce data interference, each time and for each browser, a different VM is used to capture the samples. Figure 1 depicts the research technique in the form of a flowchart.

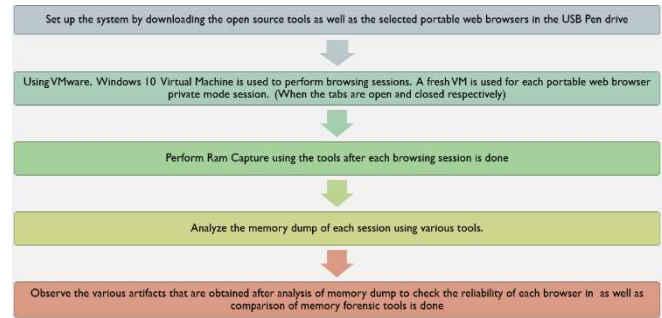


Fig. 1. Proposed Methodology for Identification of web activity traces

The Volatility Framework is a free and open set of tools for extracting digital artifacts from volatile memory (RAM) samples, written in Python and licensed under the GNU General Public License [18]. Law enforcement, military, academic, and commercial investigators all throughout the globe rely on volatility as a critical digital investigative tool. Table II elaborates the plugins used in volatility framework to identify potential evidence left behind the browsing activities.

TABLE II. PLUGINS USED IN VOLATILITY FRAMEWORK

Name of the plugin	Description
Imageinfo	To Identify information for the image
Kdbgscan	Search and dump potential KDBG values
Pslst	Print and lists all the running processes by following the EPROCESS lists.
Pstree	Print process list as a tree / hierarchical manner
Cmdline	To list and Display process command-line arguments
Envvars	To list and display process environment variables
Dlllist	Print and lists all the loaded dlls for each process
Netscan	Scans a Vista (or later) image for connections and sockets to get network information like IP addresses
Yarascan	Scans the process or kernel memory with Yara signatures from the image
Prefetchparser	Scans for and parses potential Prefetch files from the image

V. RESULTS AND DISCUSSION

It was observed during experiments that results vary based on browsers used for internet access and tools used for collection of memory dump. For all the browsers, pslst plugin was able to identify the process of respective browser even if it was closed. Yarascan is best known for its malware detection and categorization tool, but it's not limited to malware analysis. As a result, it was a preferred plugin in this study since it allows to create rules for certain strings that can be exposed as a result. Hence, it was employed to locate browsing actions on the system. Yarascan plugin was able to retrieve the sites that were accessed and few keywords used to search during browsing sessions. Table III represents the results of memory samples collected following internet activities analysed with yara rules under the volatility framework.

Post analysis it was deduced that more data can be retrieved when the portable browser's tabs are open compared

to when the tabs are closed after browsing session. The study revealed that, when google was used as search engine, more information can be obtained from memory analysis whereas privacy was well protected when duckduckgo was used as search engine. Http sites were used to check whether log in details were recoverable, however, the findings were contradictory after analysis. Information with respect to date, time, location, name of the executed file and keywords was easily retrieved by memory analysis using volatility framework. Analysis was performed by collection of memory dump at two different instances: when the browsing tabs were closed and when the browsing tabs were open.

A. When Browser Tab is Closed

For Brave portable browser, while analyzing RAM dump obtained using DumpIt, Belkasoft and Magnet using the plugin pslist and pstree, it was found that, when the process brave.exe was started, information like date, time, process id (pid 5164), and parent process id was captured, thus we got to know that brave web browser had been accessed. Only process id and name of the process was observed when the plugin cmdline, dlllist, envvars, and netscan plugin was used, no characteristic/negligible output from observed. RAM dumps obtained from DumpIt and Magnet RAM Capture retrieved details about the browsing session partially that is only facebook and gmail whereas Belkasoft Ram Capturer retrieved details about youtube and facebook..

For Tor portable browser, the process was found to be firefox.exe, where information like date, time, process id (pid 1396), parent process was shown when pslist and pstree plugin was run. cmdline and dlllist showed only the process and its id whereas no output was shown from netscan and envvars. Only RAM dump obtained from Belkasoft Ram Capturer was able to retrieve information about facebook, other tool's ram dumps couldn't capture these remnants from the portable browser's private mode session using yarascan.

Similarly for Vivaldi, after running the commands pslist and pstree, vivaldi.exe process was found where information like date, time, parent process id, process id (Pid 2276) was found. No characteristic output was observed for cmdline, dlllist, netscan and envvars. After running yarascan plugin it was observed that memory dump captured using belkasoft RAM Capturer was able to recollect the residual artifacts of the browsing session of gmail and facebook

However, for Maxthon, there was an absence of browser specific process like "Maxthon.exe" in the list of processes that were observed after pslist and pstree plugin was run. Only process id and name of the process was observed when the plugin netscan was used. No output was observed in envvars, cmdline and dlllist plugins. RAM dumps obtained from DumpIt and Magnet RAM Capture retrieved details about one particular browsing session that is only facebook whereas Belkasoft Ram Capturer retrieved details about gmail and facebook.

B. When Browser Tab is Open

During analysis of memory dumps of live browser sessions obtained from the tools mentioned above we observed that number of processes were more as well as multiple instances of the browser sessions were recorded in the output obtained from pslist and pstree plugins for all the browsers.

For Brave portable browser it was observed that when the plugin cmdline, was run it showed the file path of the browser "E:\ Brave-portable\app\brave.exe" and name of the process whereas in dlllist the same was observed along with the date and time of the process and dll files of the executable files in all the memory dumps. Output for envvars was only obtained from the memory dump captured by DumpIt which enlisted all the environment variables like name of the users, desktop name etc. netscan plugin listed the processes, date and time but it wasn't able to retrieve the type of connection and ip address.

For Tor when the plugin cmdline, was run it showed the file path of the browser "E:\TorBrowser\Browser\firefox.exe" and name of the process tor.exe and firefox.exe was observed. In dlllist the same was observed along with the date and time of the process. DLL files of the executable files were not captured in all the memory dumps. There was no output for envvars for all the memory dumps. Netscan plugin listed two processes, tor.exe (Pid 6664) and obfs4proxy.exe (Pid 5096) which showed "LISTENING" connections to non-standardized ports as output.

For Vivaldi it was observed that when the plugin cmdline, was run it showed the file path of the browser "E:\ vivaldi open\Application\vivaldi.exe" and name of the process. In dlllist files related to vivaldi.exe was shown, DLL files(.dll) were stored in the external storage drive in all memory dumps. Output for envvars was not retrieved. netscan plugin could not to retrieve the type of connection and ip address.

For Maxthon it was observed that when the plugin cmdline, was run it showed the file path of the browser "E:\maxthon_portable_6.1.0.2000x64\MaxthonPortable\Maxthon.exe" and name of the process whereas in dlllist the same was observed along with the date and time of the process and dll files of the executable files in all the memory dumps. Using envvars plugin we got the information of number of CPUs installed as well as the hardware architecture, the process's current directory, temporary directory, session name, computer name, username, and other valuable artifacts on the process Maxthon.exe. netscan plugin listed the one process Maxthon.exe (pid 6116) which was found to be "LISTENING" connections to non-standardized ports.

Yarascan plugin was able to retrieve all the browsing sessions for the memory dumps obtained from all the RAM capturing tools (DumpIt, Belkasoft and Magnet) for only three browsers (Brave, TOR and Vivaldi). However, in maxthon, details about amazon, cybercrime, gmail and youtube was only retrieved. There were no traces regarding facebook while the session was live but traces were found in the terminated session. Prefetchparser was able to fetch the prefetch files but it was encrypted in all the memory dumps.

TABLE III. LIS OF ALL EVIDENCE OF WEBSITE VISIT AND KEYWORDS

Browsers	Keyword analysis using yarascan plugin				
	Amazon	Facebook	Cybercrime	Gmail	YouTube
Brave	=amazon.c...om, p,https://image s-na.ssl-L.W... /...I/51Nfltd59 mL_RC(01oxUylwj HL.cs...s_l?AU IClients/Brila.W .C...U.www.a&.. ...Samsung-Gala xy-Note-10-Unloc ked/...dp/B07Z3	=facebook&oq=fac ebook&aqs=chrome	=cybercrime&sour ce=lnms&t...bm=	=gmail&em=resear chergal23%40gmai l.com..L.)..").	=yt&cp=22&gs_id= 3f&q=how%20to%20 crack%20password &hr=t&xsc'...%
TOR	=amazon.c...om, p,https://image s-na.ssl-L.W... /...I/51Nfltd59 mL_RC(01oxUylwj HL.cs...s_l?AU IClients/Brila.W .C...U.www.a&.. ...Samsung-Gala xy-Note-10-Unloc ked/...dp/B07Z3	=facebook.com...	=cybercrime&tax= images&ta=images &lat=https%3A%2F %2Fwww.nwrocu.po lice.uk%2Fmedia% 2Fs--043f8a5d2d3 8b86fd7067d3f5ff bcc15%2Fmedia%2F 0r1JgBNqfGS1Kjd2 weWxq1cWRN9C0p9 z9Iwewct.jpeg#..	=gmail&ust=..917 945..S.....g= AFQjCNFEDt3vU06C PTX_NHwsN7LX5... Zmwogg...\$https: //www..url?q=... .S..A...@.Choo ser?E...%3Dresear chergal23@z...%, 22...&gs_id=3f&q=	=youtube&ds=yt&c p=31&gs_id=3d&q= how%20to%20crack %20password%20us ing%20cmd&xhr=t& xssi=t.....
Vivaldi	=amazon.in;pt=De tail;...slot=.. ;pid=B08M6HMGQN; q=Samsung....no te.10;ar..3ca0e7	=facebook.*.... X....M...\\....I	=cybercrime&sour ce=lnms&tbm=lsch &sa=X&ved=2ahUKE wjBhMDPorryAhVBz TgGHwQIDuAQ_AUoA noECAEQBA&b1w=18 B&abth=809#lmgrc =uuES4KL6fOF3GM" "thumbnail": "da ta:image/jpeg;ba	Q...https://www. google.com/url?q =https://account s.google.com/acc ountchooser?Email =l33dresearcherga l23@gmail.com&x26 continue%3Dhttps	=youtube&ds=yt&c p=22&gs_id=2n&q= how%20to%20crack %20passwords&xhr
Maxthon	=amazon....us;p t=Detail;slot=ad f.;pid=B07WF.... HPZ2D;q=samsung. note.10;ar..123b	=facebook&client =aff-maxthon-max	=cybercrime&clie nt=aff-maxthon=.	=gmail_message;d url=htx;...hd f.&/dig8;/be.%e- a/;....15...5ac counts.g...oogl e.com/A..ChooSer ?EA.=researche.. h.rgal23@gm...q.&	=youtube&ds=yt&c p=24&gs_id=36&q= how%20to%20crack %20password%20cm

VI. CONCLUSION

This research sought to investigate the potential evidence left over after the activities carried out through private mode in portable browsers. From the output it can be concluded that almost all browser's browsing details can be analyzed through volatile memory analysis. The portable web browsers in private mode can retain information like which portable browser was used, date and time, websites that were visited and many more. The results can be found more if the browser window is open as compared to the closed window. Not all plugins were able to retrieve browsing artefacts from the memory dump, indicating that the tools used for acquisition mattered. Windows 10 didn't support several plugins. Volatility plugins and extra plugins produced an output in normal mode surfing, however, the data could not be recovered in private mode browsing using a portable browser, yet yarascan could identify the domains and keywords used for online activities. These findings can be used as a starting point for looking at a variety of portable browsers, including TOR.

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